Glossary

Every new word in Level 1, in A to Z order

Tip: How to use this glossary

Words are in alphabetical order. The gray label tells you which week you first meet the word (W1 = week 1, B1 = buffer session 1). Words in `this font` are typed exactly like that in Python. Found a word that is not here? Write it on the last page with your own meaning!

A

accessibility W5

Making things that everyone can use, including people who see colors differently or need bigger text.

Example: Using color-blind-safe colors and also different shapes.

algorithm W1

A list of exact steps, in the right order, that solves a problem or does a job.

Example: A recipe for a sandwich.

angle W2

How far you turn, measured in degrees. A full turn is 360 degrees. A shape with n sides turns $360 \div n$ at each corner.

Example: A square turns 90 degrees at each corner.

B

backdrop W1

The picture behind everything on the Scratch stage.

broadcast W5

A Scratch message sent to every sprite. Sprites with a matching “when I receive” block start running.

Example: “broadcast start game”.

bug W1

A mistake in a program that makes it do something we did not want. Bugs are normal: every programmer meets them every day.

Example: The turtle turns `90` instead of `120`.

C

call W10

To use a function, by writing its name with brackets.

Example: `house(0, 0, ORANGE)` calls the function `house`.

color code W9

A color written as # and six letters or numbers, so the computer knows exactly which color you mean.

Example: `"#0072B2"` is blue.

comment W7

A note for humans in the code. Python ignores it. In Python it starts with #.

Example: `# draw the roof`

comparison W9

A question that is True or False, made with $>$ (bigger), $<$ (smaller), == (equal) or != (not equal).

Example: `stars > 5`

conditional W3

Code that only runs if something is true. Also called an **if**.

Example: “If touching the edge, then bounce.”

coordinates W1

Two numbers, x and y, that give a place on the screen. x is left–right, y is down–up. The middle is $(0, 0)$.

Example: `t.goto(200, 150)`

costume W2

One of the pictures a Scratch sprite can wear. Switching costumes quickly makes animation.

credit W4

Saying who made the art, sounds or ideas you used. It is fair and kind.

Example: “Cat sounds from the Scratch library.”

D

debug W1

To find a bug and fix it. A good strategy: read the error, check the last thing you changed, test one small part at a time.

decomposition W5

Breaking a big problem into smaller parts that are easier to solve.

Example: A house = walls + roof + door + windows.

`def` W10

The Python word that starts a new function (your own command).

Example: `def square(size):`

driver W1

In pair programming, the person using the mouse and keyboard.

E

`elif` W9

Short for "else if": another question to check when the first `if` was false.

Example: `elif n == 10:`

else W3

The part of an if that runs when the question is false.

error message W7

The red text Python shows when it cannot run your code. It tells you the line number and a clue.

Example: `NameError: name 'Print' is not defined`

event W3

Something that happens and starts code running.

Example: Clicking the green flag, pressing a key.

extension W2

Extra blocks you can add to Scratch, like the Pen.

F

feedback W3

Helpful comments about someone's work. We use "two stars and a wish": two things you like, one idea to improve.

`for` loop W8

Python's repeat block. The lines pushed in underneath it are repeated.

Example: `for i in range(4):`

forever W2

A Scratch loop that never stops (until you click the red stop sign).

function W10

A named set of steps you can use again and again. Python's version of a My Block.

Example: `square(50)`

G

green flag W1

The Scratch button that starts a project.

I

if W3

A block or line that asks a question and only runs its steps when the answer is true.

Example: `if mood == "happy":`

indentation W8

Spaces at the start of a line (4 in Python). They show which lines are inside a loop, an if or a function.

input W9

Information that goes into a program, often typed by the user. In Python `input()` asks a question and gives back the answer as text.

Example: `answer = input("How many? ")`

integer (`int`) W9

A whole number, like 3, 0 or −7. `int()` turns text like `"5"` into the number 5.

Example: `n = int(answer)`

L

loop W2

Code that repeats. Loops save you writing the same steps again and again.

Example: repeat 4, `for i in range(4):`

M

My Block W5

A block you make yourself in Scratch with “Make a Block”. It can have inputs.

Example: “define flower (size)”

N

navigator W1

In pair programming, the person who reads the instructions, thinks ahead and spots bugs.

nested loop W8

A loop inside another loop.

Example: Repeat 6 times: draw a square.

O

output W7

What a program shows or does: text, a drawing, a sound.

P

pair programming W1

Two people coding together on one computer: a driver and a navigator who swap roles.

parameter W5

An input that a My Block or function needs. It makes the block flexible.

Example: In `square(size)`, `size` is a parameter.

pen W2

Lets a sprite or turtle draw a line where it moves. Pen up: no line. Pen down: draw.

pixel B1

One tiny square of color. Screens make pictures out of millions of pixels.

playtest W11

Letting someone try your program so you can see what works and what confuses them.

`print` W7

The Python command that shows text in the Shell.

Example: `print("Hello!")`

privacy W1

Keeping personal information (real name, address, school, photos, passwords) safe. We use nicknames.

program W1

A set of instructions a computer follows.

Python W7

A programming language where you type words instead of dragging blocks.

R

random W4

Picked by chance, so you cannot guess it. Great for surprises and games.

Example: `random.randint(1, 6)`, "pick random 1 to 6"

`range` W8

Tells a `for` loop how many times to repeat.

Example: `range(5)` repeats 5 times.

remix W4

Making something new from someone else's work, and giving them credit.

repeat W2

A Scratch loop that runs the blocks inside a set number of times.

run W1

To start a program. In Thonny, the green Run button or F5.

S

script W1

A stack of Scratch blocks that are joined together.

sensing W3

Scratch blocks that check things, like "touching edge?" or "key space pressed?".

sequence W1

Steps that happen one after another, in order. Order matters!

Shell W7

The part of Thonny at the bottom where `print` shows text and you type answers.

sprite W1

A character or object in Scratch that you can program.

Example: The Scratch cat.

stage W1

The area in Scratch where the sprites move and the project plays.

string W9

Text in a program, written inside quotes. `input()` always gives a string.

Example: `"day"`, `"25"`

syntax W7

The spelling and grammar rules of a programming language.

syntax error W7

An error because the code breaks Python's rules, like a missing bracket or colon.

Example: `print("hi"`

T

test W7

Running a program on purpose to check that it works. `check_my_work.py` runs tests for you.

Thonny W7

The app we use to write and run Python.

turtle W7

A Python drawing tool: an arrow you move around the screen that draws lines.

Example: `t = turtle.Turtle()`

V

variable W4

A labeled box that remembers a value, like a score or an answer. The value can change.

Example: `score = 0`

More words from your workbook

accessible Easy for everyone to use, including people who see or hear differently. (week 5)

bug / debugging A mistake in a program / finding and fixing it. (week 1)

checkpoint A short check that shows what you know and what to practice next. (week 6)

debugging Finding and fixing bugs, step by step, like a detective. (week 6)

Demo Short for demonstration: running your program and showing how it works. (week 12)

driver / navigator The driver clicks; the navigator reads, checks and helps. (week 1)

Editor The top part of Thonny where you write your program. (week 7)

Feedback card A card with two stars and a wish for an artist. (week 12)

if / elif / else Python's way to choose: "if this, else if that, otherwise..." (week 9)

if-else Runs one set of blocks for yes and another for no. (week 3)

Iterate To improve something by going round the build–run–check loop again and again. (week 11)

Milestone A checkpoint on the way to finishing a project. (week 11)

Off-by-one bug When a loop runs one time too many or one time too few. (week 8)

Reflection Thinking back about what you did, what you learned and what you want next. (week 12)

Showcase An event where you show your work to others. (week 12)

text code Code made of typed words and symbols. (week 6)

My own words

Add new words you meet, and write the meaning in your own words.

Word	What it means (in my words)